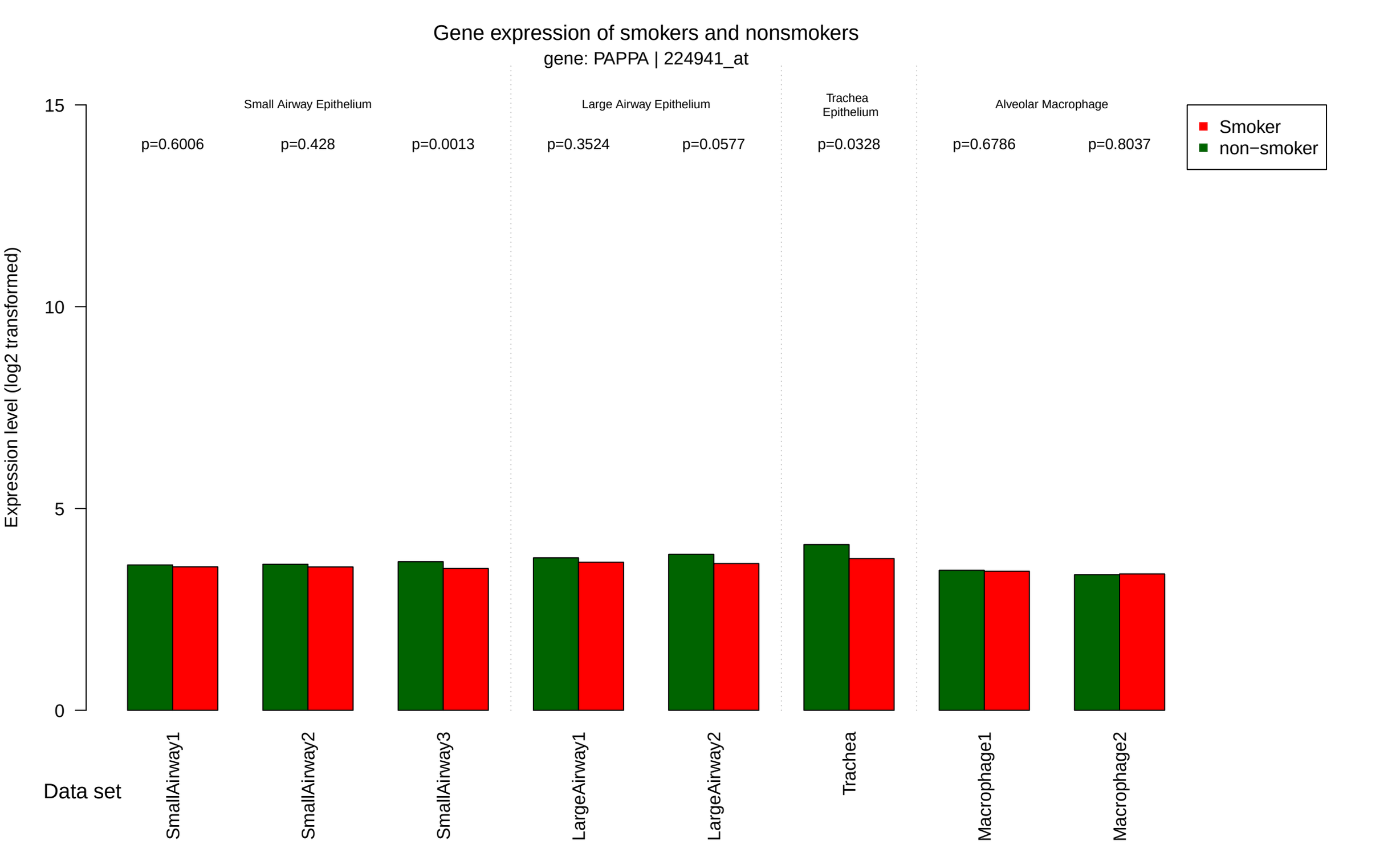
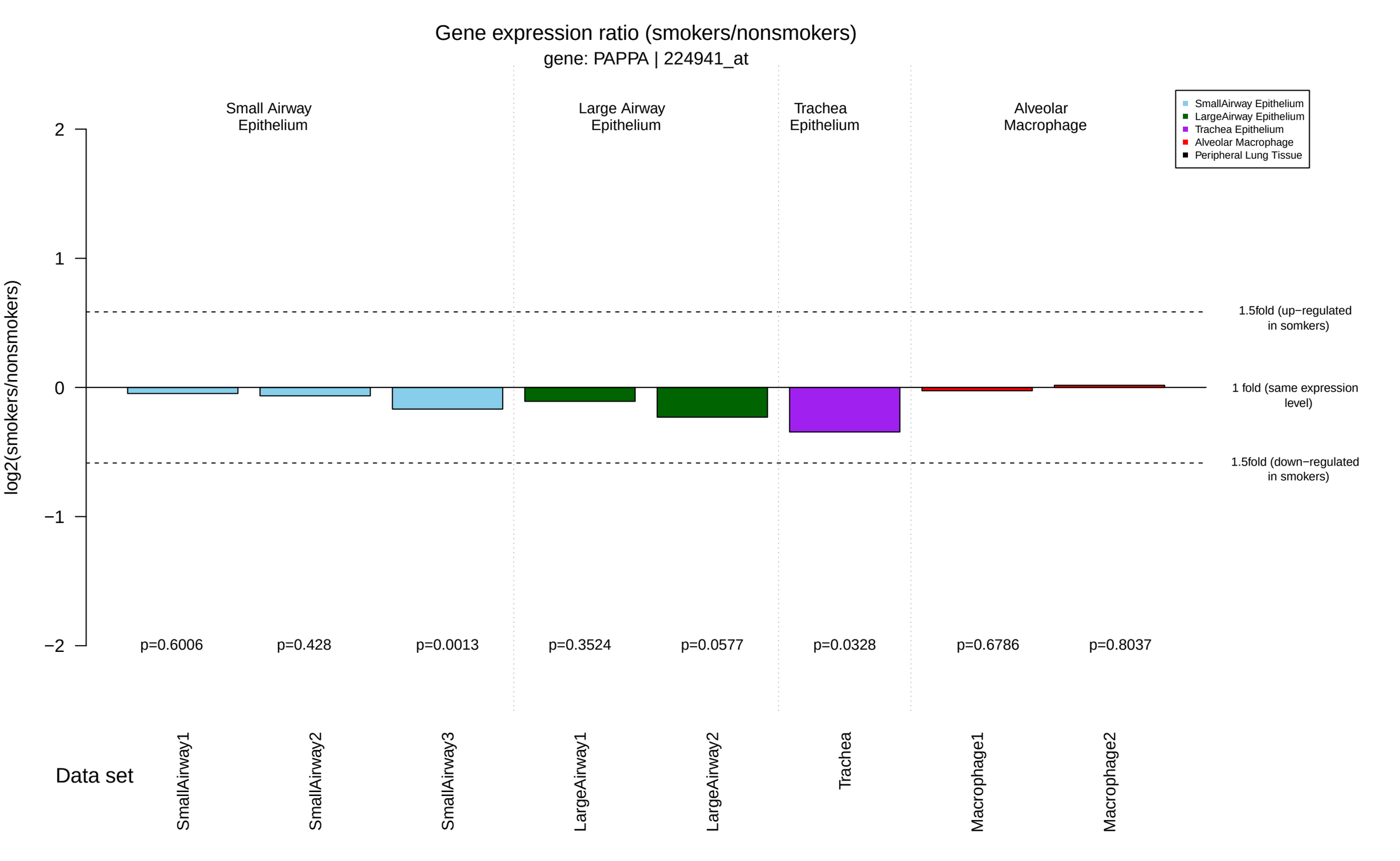
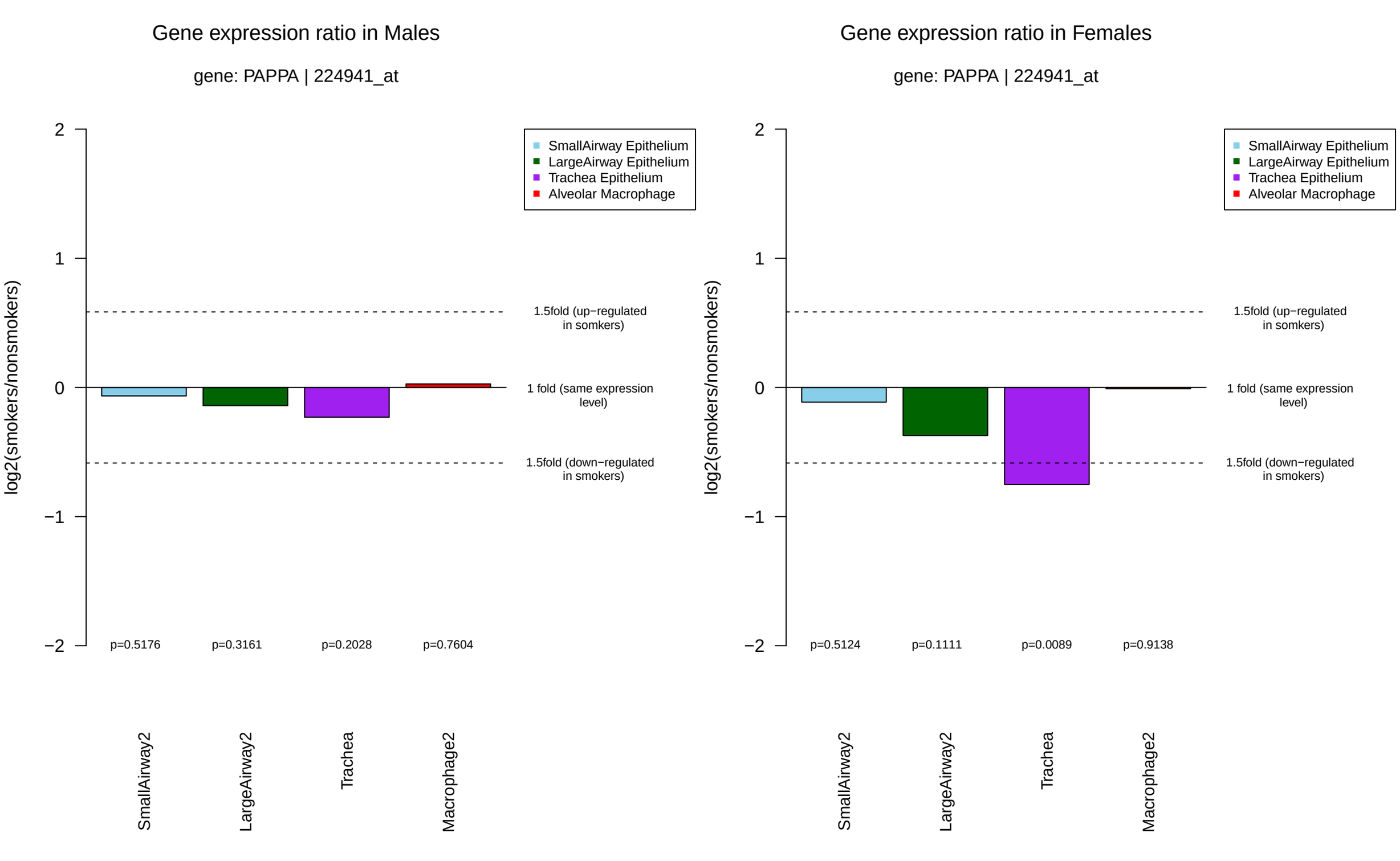
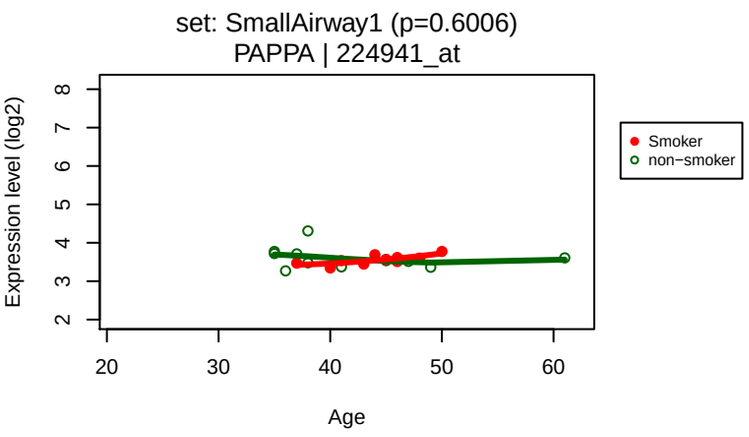
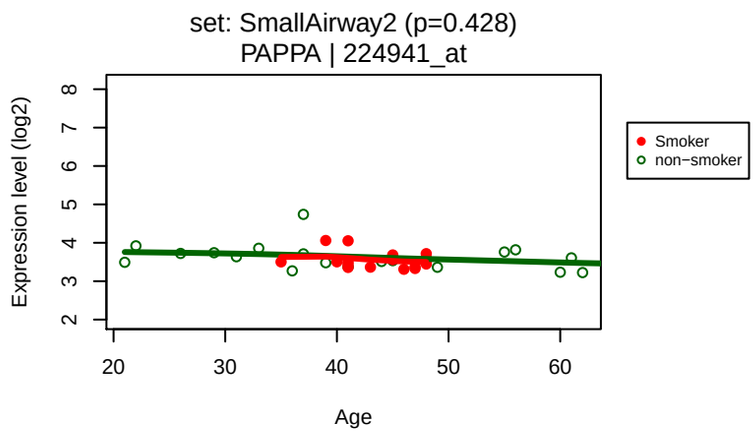
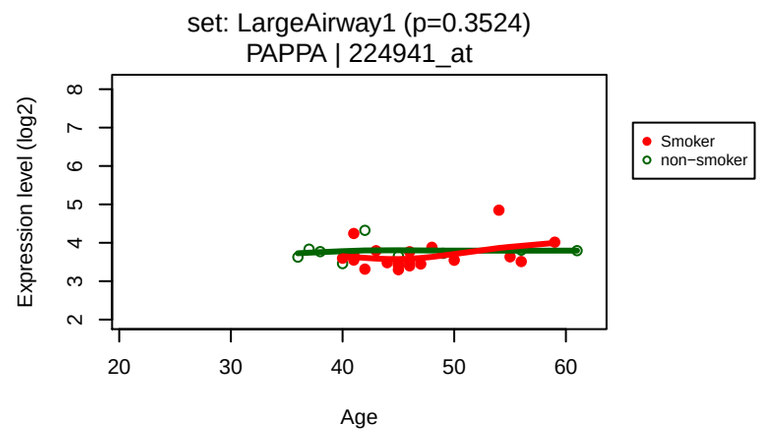

Gene expression of smokers and nonsmokers
gene: PAPPA | 224941_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6006
p=0.428
p=0.0013
p=0.3524
p=0.0577
p=0.0328
p=0.6786
p=0.8037
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

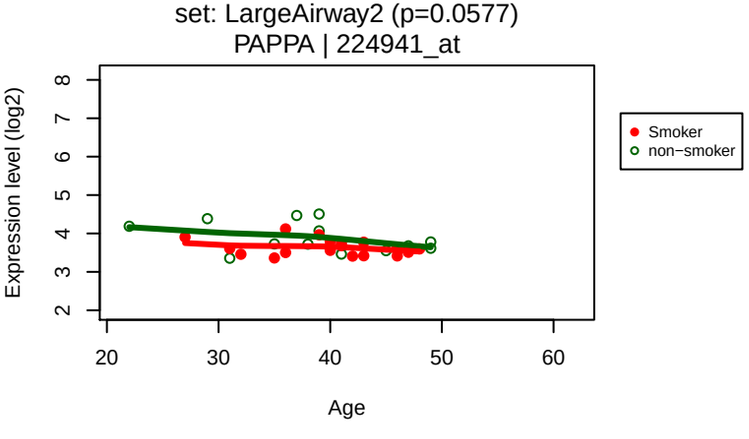

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA | 224941_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6006
p=0.428
p=0.0013
p=0.3524
p=0.0577
p=0.0328
p=0.6786
p=0.8037
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

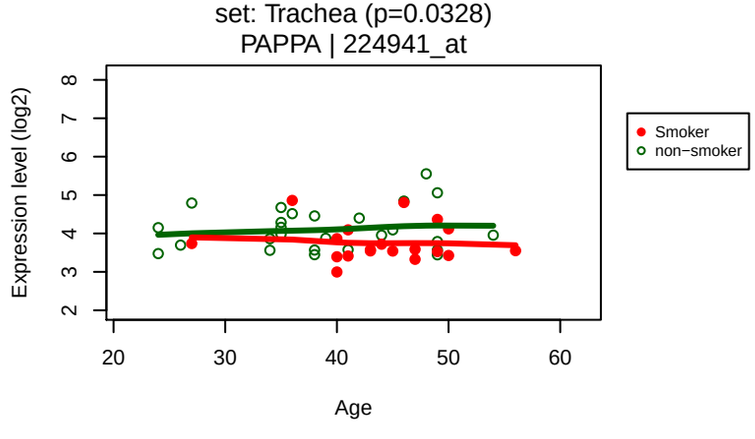

Gene expression ratio in Males
gene: PAPPA | 224941_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5176
p=0.3161
p=0.2028
p=0.7604
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA | 224941_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5124
p=0.1111
p=0.0089
p=0.9138
SmallAirway2
LargeAirway2
Trachea
Macrophage2

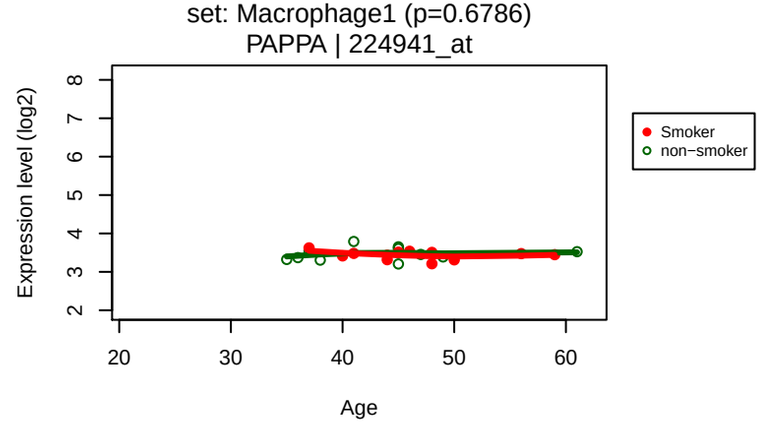

set: SmallAirway1 (p=0.6006)
PAPPA | 224941_at
Expression level (log2)
Age
Smoker
non−smoker

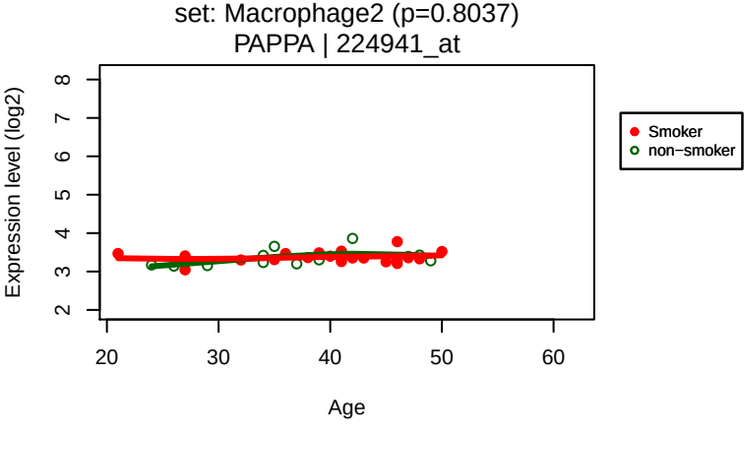

set: SmallAirway2 (p=0.428)
PAPPA | 224941_at
Expression level (log2)
Age
Smoker
non−smoker

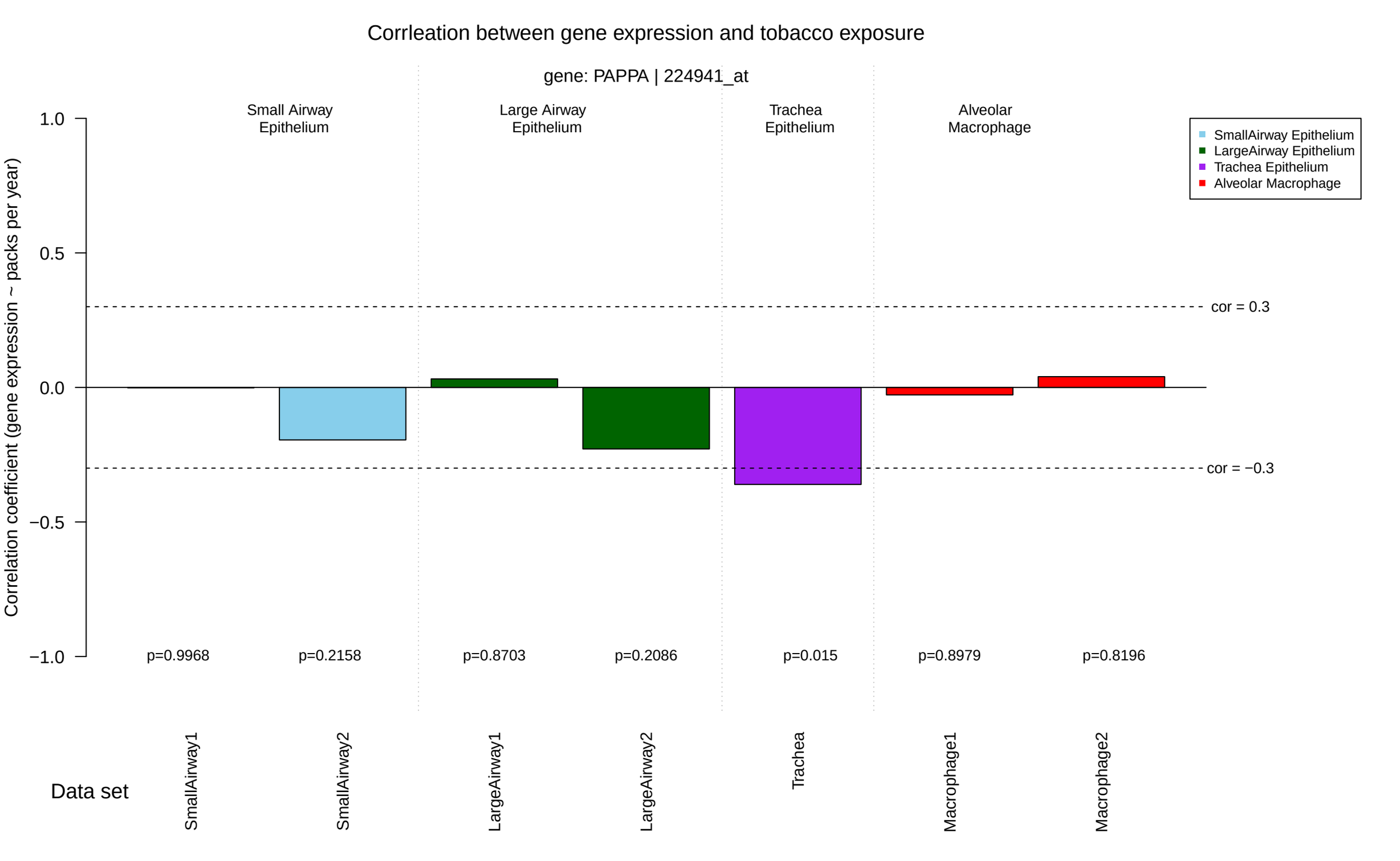

set: LargeAirway1 (p=0.3524)
PAPPA | 224941_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0577)
PAPPA | 224941_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0328)
PAPPA | 224941_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6786)
PAPPA | 224941_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8037)
PAPPA | 224941_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA | 224941_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9968
p=0.2158
p=0.8703
p=0.2086
p=0.015
p=0.8979
p=0.8196
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2